

NASA SVS Visualization – 30728 (Liu et al., 2014)

Link: <https://svs.gsfc.nasa.gov/30728>

This entry highlights NOAA's Coral Reef Watch (CRW) advancements in satellite-based coral bleaching monitoring. The associated Liu et al. (2014) paper introduced global 5-km resolution thermal stress products—HotSpots and Degree Heating Weeks (DHW)—derived from satellite SST data. These provide reef-scale insight into thermal stress, supporting both global comparisons and localized alerts.

Key Contributions:

- Pioneered reef-scale global products at 5-km resolution.
- Enhanced accuracy of bleaching detection compared to coarser SST.
- Gave managers globally consistent tools (HotSpot, DHW) for risk tracking.

Limitations:

- SST proxies do not capture all bleaching drivers (e.g., light stress, local hydrodynamics).
- Requires local field validation for finer precision.

Overall, the paper shows a critical step in transitioning coral reef monitoring from research-grade to operational, laying groundwork for higher-resolution and predictive products.

Paper Title	Key Findings	Limitations	Year
Reef-Scale Thermal Stress Monitoring of Coral Ecosystems: New 5-km Global Products from NOAA Coral Reef Watch	Introduced 5-km global coral bleaching monitoring products (HotSpot & DHW) using satellite SST data. Improved reef-scale monitoring accuracy.	Dependent on satellite SST proxies; limited local reef validation.	2014

NOAA CRW Four-Month Coral Bleaching Outlook (CFS-based)

Link: https://coralreefwatch.noaa.gov/satellite/bleachingoutlook_cfs/index.php

This NOAA CRW resource details the Coral Bleaching Outlook system, which predicts thermal stress up to four months ahead. Two major studies underpin the system:

- Eakin et al. (2012): Introduced the first probabilistic forecasts using CFSv1. Demonstrated

that climate forecasts could be linked with CRW products to provide seasonal-scale warnings.

- Liu et al. (2018): Enhanced the system using CFSv2, producing higher resolution, ensemble-based probabilistic forecasts (Outlook v5.0). Provided operational global maps with early warning value.

Key Contributions:

- Created the world's first global seasonal bleaching outlook.
- Enabled managers to prepare for bleaching events months ahead.
- Established probabilistic methods in reef climate forecasting.

Limitations:

- Forecast skill drops after 20 weeks.
- Regional performance varies due to model biases and local complexity.

Overall, the Coral Bleaching Outlook bridges climate prediction and reef conservation, offering a proactive approach to management.

Paper Title	Key Findings	Limitations	Year
Ghost of bleaching future: Seasonal Outlooks from NOAA's Operational Climate Forecast System	Introduced probabilistic Four-Month Outlook with CFSv1; showed seasonal early warning feasibility.	Lower resolution, fewer ensemble members, early model limitations.	2012
Predicting Heat Stress to Inform Reef Management: NOAA Coral Reef Watch's 4-Month Coral Bleaching Outlook	Documented Outlook v5.0 using CFSv2; combined SST forecasts with HotSpot/DHW thresholds for global probabilistic forecasts.	Forecast accuracy declines with longer lead times; varies regionally.	2018

Automated Detection of Coral Bleaching Using YOLOv8 (Patel et al., 2024)

Link: [Uploaded PDF](#)

This 2024 study by Patel et al. presents a deep learning-based approach for detecting bleached corals using YOLOv8. A dataset of ~1500 annotated images (healthy vs. bleached) was used to train a YOLOv8 small model. Results showed 94% accuracy for healthy coral detection and 79% for bleached corals.

The study also developed a GUI using PySimpleGUI, which lets users upload coral reef

images and instantly see bounding boxes and bleaching predictions. This real-time usability feature makes the approach practical for non-technical users.

Key Contributions:

- Demonstrated state-of-the-art deep learning in reef ecology.
- Real-time GUI tool improves accessibility.
- Pioneers automation of coral reef health monitoring.

Limitations:

- Dataset size was relatively small and augmented.
- Lower accuracy for bleached corals.
- Needs validation across reef types and environmental conditions.

Overall, this paper highlights the promise of AI-driven monitoring as a scalable, proactive conservation tool.

Paper Title	Key Findings	Limitations	Year
Automated Detection of Coral Bleaching Using YOLOv8: A Deep Learning Approach for Reef Health Monitoring	YOLOv8 model detects bleached vs healthy corals; achieved ~94% accuracy for healthy, ~79% for bleached; GUI built for real-time detection.	Dataset small (~1500 images), class imbalance, lower accuracy for bleached; generalization risks.	2024 (Dec)

NOAA CRW Near-Real-Time Global Monitoring (Liu et al., 2004/2005)

Link: [ResearchGate PDF](#)

This paper (Liu, Strong, Skirving, Arzayus, 2004/2005) describes the establishment of NOAA's Coral Reef Watch (CRW) as the first operational near-real-time satellite monitoring system for coral bleaching. Key metrics included HotSpots (SST anomalies above climatology) and Degree Heating Weeks (cumulative heat stress).

CRW products were made freely available online and updated twice weekly, offering unprecedented global coverage for reef managers.

Key Contributions:

- Standardized global bleaching stress metrics.
- Demonstrated early warnings during the 2002 Hawaii and GBR events.
- Marked the transition to operational, near-real-time reef monitoring.

Limitations:

- ~50 km resolution missed local reef variability.
- Limited field validation at the time.

Future Enhancements (outlined in the paper):

- Higher spatial resolution.
- Additional products (short-term trends, stress duration).
- Email alerts and automated dissemination.
- Integration with predictive models.

Overall, this work laid the foundation for all subsequent CRW monitoring tools.

Paper Title	Key Findings	Limitations	Year
Overview of NOAA Coral Reef Watch Program's Near-Real-Time Satellite Global Coral Bleaching Monitoring Activities	Launched global near-real-time bleaching monitoring (HotSpots, DHW, SST anomalies); supported managers during 2002 GBR and Hawaii events.	50 km coarse resolution; limited field validation.	2004/2005

Literature Survey (from YOLOv8 paper)

Link: Cited works in 83.pdf

The YOLOv8 study also surveyed prior works applying deep learning and computer vision to coral health monitoring. Five notable papers were discussed:

1. Bautista-Hernández et al. (2022): Applied CNNs for coral disease detection (SCTLD) with 94% accuracy on 1,555 images.
2. Mahmood et al. (2016): Pioneered automatic annotation of large-scale AUV coral imagery using CNNs, applied in Western Australia.
3. Jamil et al. (2021): Proposed Bag of Features (BoF)-based deep learning for bleached coral detection; achieved 99% accuracy.
4. Thamarai & Aruna (2023): Used CNNs (ResNet50, InceptionV3, custom models) for stressed coral identification; achieved up to 90% accuracy.
5. Corruz et al. (2021): Developed a mobile CNN-based app for bleaching classification on Android devices; achieved ~93% accuracy.

Together, these studies show the progression from manual annotation to automated, mobile-ready deep learning tools for reef monitoring.

Paper Title	Key Findings	Limitations	Year
Coral Reef Disease and Bleaching Identification (Bautista-Hernández et al.)	CNN achieved ~94% accuracy detecting SCTLD disease in Caribbean corals using 1,555 images.	Dataset small; specific to disease type; limited bleaching focus.	2022
Automatic Annotation of Coral Reefs Using Deep Learning (Mahmood et al.)	CNN automated large-scale annotation of AUV images in Western Australia; advanced ecological monitoring.	Early CNN limitations; still required careful tuning and validation.	2016
BoF-Based Deep Learning for Bleached Coral Detection (Jamil et al.)	Hybrid deep CNN+BoF features; achieved 99% accuracy in bleaching detection and localization.	High reported accuracy but limited generalization studies.	2021
Stressed Coral Reef Identification with CNNs (Thamarai & Aruna)	Used ResNet50, InceptionV3, and custom CNN; up to 90% accuracy for stress classification.	Accuracy varied; pretrained models underperformed; dataset limited.	2023
BahurApp: Mobile Coral Bleaching Monitoring (Corruz et al.)	CNN-based mobile app on Android; ~93% accuracy for bleaching classification; GPS integration.	Performance sensitive to dataset quality; limited smartphone camera resolution.	2021

International Journal of Design & Nature and Ecodynamics

Title: *Forecast Coral Bleaching by Machine Learnings of Remotely Sensed Geospatial Data*

Authors: Tanatpong Udomchaipitak, Nathaphon Boonnam, Supattra Puttinaovarat, Paramate Horkaew

Published: *International Journal of Design & Nature and Ecodynamics*, Vol. 17, No. 3, June 2022i

This study focuses on predicting coral bleaching using **machine learning (ML) models** and **remotely sensed geospatial data**. Unlike earlier works that mostly relied only on sea surface temperature (SST), this paper integrates multiple environmental variables—**SST, turbidity, salinity, and wind speed**—to improve accuracy.

The authors compared four ML algorithms:

- **Random Forest (RF)**
- **Decision Tree (DT)**
- **Artificial Neural Network (ANN, MLP & RBF variants)**

Among these, **Random Forest proved most effective**, achieving up to **97.25% accuracy in cross-validation** and **88.24% accuracy on unseen test data**.

A key contribution is the creation of an **interactive web-based application**, which visualizes coral bleaching forecasts on maps. This tool allows environmental agencies to monitor bleaching events more efficiently without depending solely on costly and time-consuming field surveys.

Title	Key Findings	Limitations	Year of Release
Forecast Coral Bleaching by Machine Learnings of Remotely Sensed Geospatial Data (Udomchaipitak, Boonnam, Puttinaovarat & Horkaew)	Used multiple geospatial factors (SST, turbidity, salinity, wind speed) to forecast coral bleaching. - Compared four ML models: ANN (MLP), Random Forest (RF), Decision Tree (DT), ANN (RBF). - RF performed best with 97.25% accuracy (10-fold CV) and 88.24% accuracy (unseen data), Kappa = 0.83. - Developed an interactive web-based visualization tool to map bleaching forecasts for decision-making.	- Relies only on environmental parameters from remote sensing; does not include human activities (e.g., pollution, coastal development). - Accuracy depends on quality and resolution of satellite data (normalized via Kriging). - Limited to case study areas (Thailand); generalization needs testing elsewhere.	2022